# Distributed Image Classification with Federated Learning

Bachelor Thesis Project

Milica Jankov
Supervisor: Prof. Carlo Fischione

# Motivation

- Modern systems generate massive amounts of **distributed data**

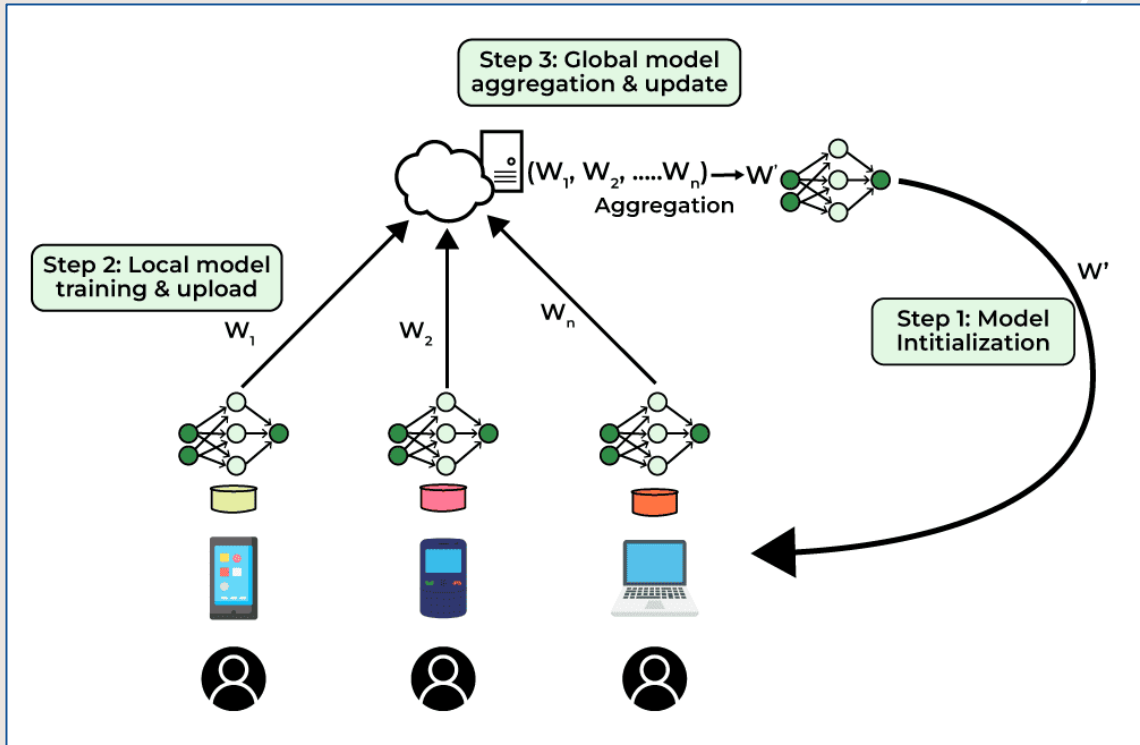
- Centralized machine learning requires:
  - Collecting all data in one place
  - High communication cost
  - Potential privacy concerns

- In many real-world scenarios, centralized training is **inefficient or infeasible**

- **Federated Learning (FL)** enables collaborative model training **without sharing raw data**

# What Is Federated Learning?

- Training is performed across multiple **distributed clients**
- Each client:
  - Keeps its local data
  - Trains a local model
- A central server:
  - Collects model updates
  - Aggregates them into a global model
- Raw data **never leaves the client**



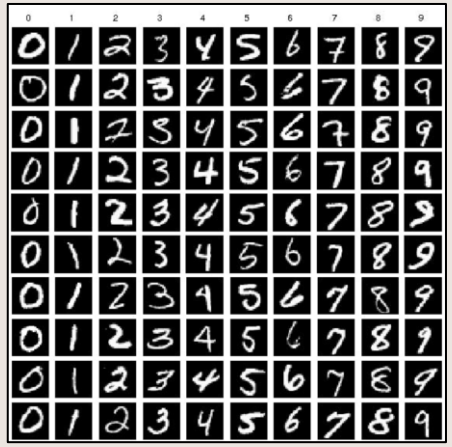
# Project Goal

The goal of this Bachelor thesis is to:

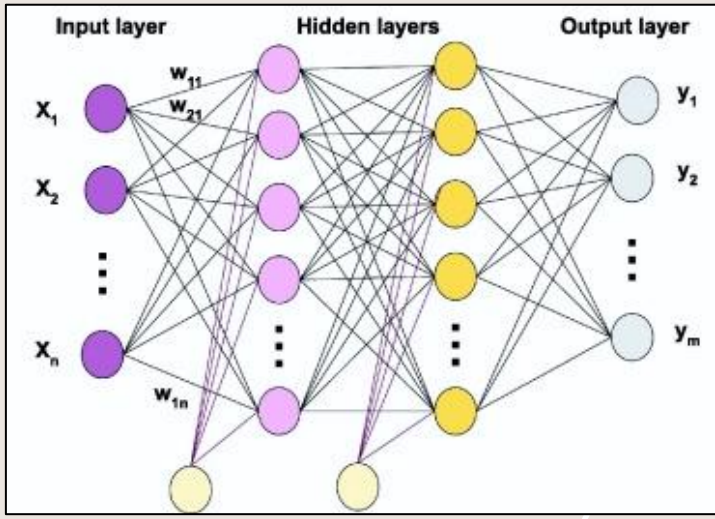
- Understand **how federated learning works in practice**
- Compare federated learning with **centralized training**
- Study how system-level factors affect learning performance

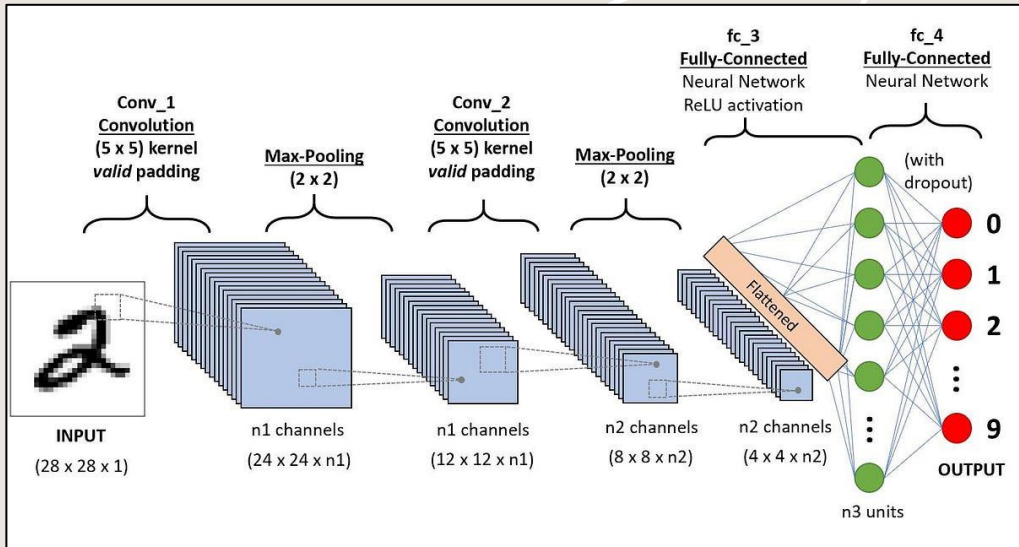
This is an **experimental and analytical project**, not algorithm design.

# Dataset and Task



- Dataset: **MNIST handwritten digits**
- Task: **Image classification**
- Model:
  - Simple neural network (MLP or small CNN)
- Evaluation metric:
  - Classification accuracy on test data



# Baseline: Centralized Learning

- Train a single model using:
  - All training data at one location
- Purpose:
  - Establish a **performance reference**
- Results to report:
  - Training and test accuracy
  - Learning curves

This baseline will be used for all comparisons.

# Federated Learning Setup

- Simulate multiple clients
- Each client:
  - Holds a subset of MNIST
  - Trains locally for several epochs
- Server:
  - Aggregates client models using **Federated Averaging (FedAvg)**
- Training proceeds in **communication rounds**

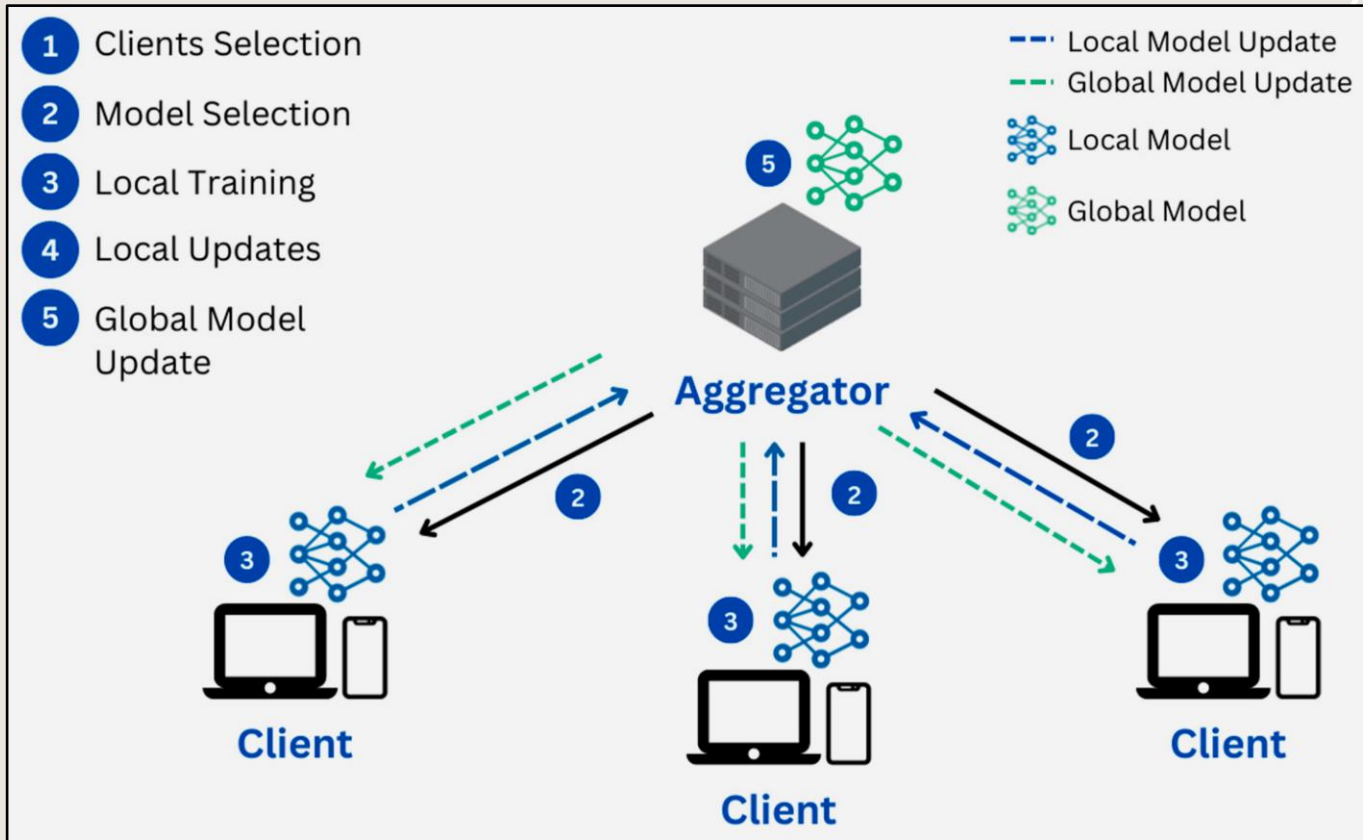
# Key Experimental Questions

The thesis focuses on answering:

- How does federated learning compare to centralized learning?
- How does performance change when:
  - The number of clients increases?
  - Data is unevenly distributed across clients?

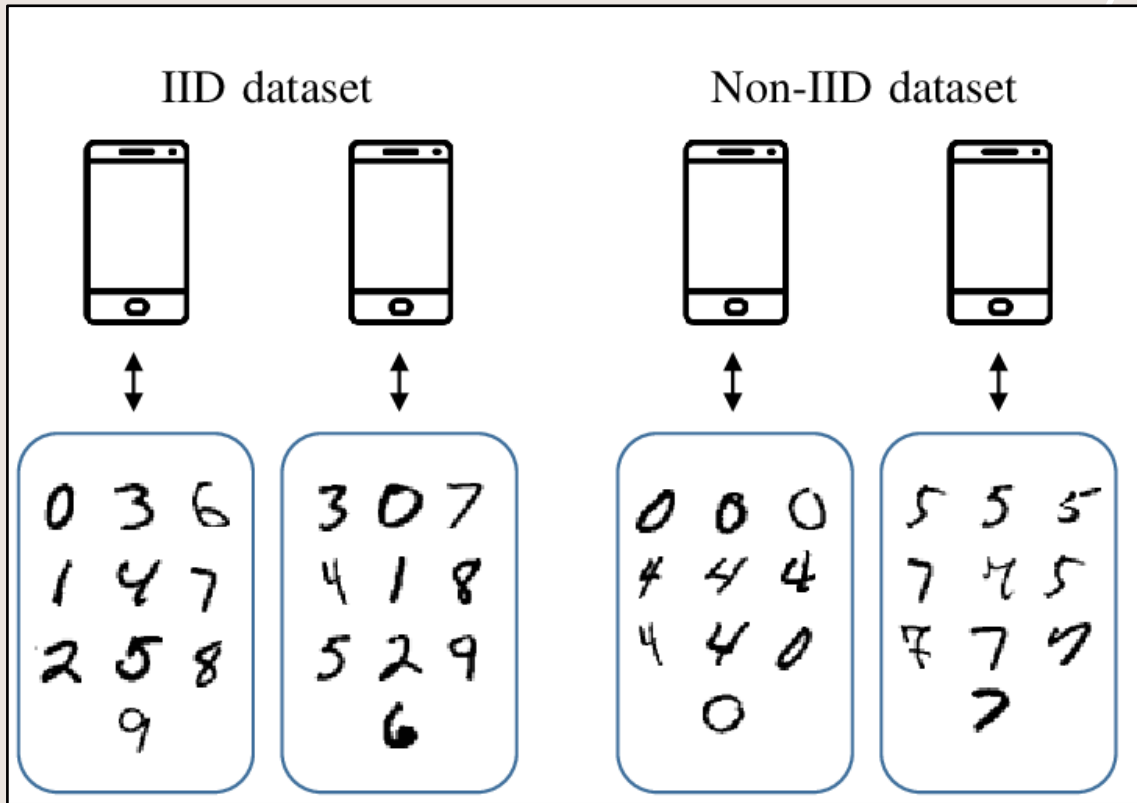
# Experiment 1: Number of Clients

- Vary the number of clients:
  - Example: 5, 10, 20, 50
- Keep total data size fixed
- Analyze:
  - Convergence speed
  - Final model accuracy
  - Stability across runs



# Experiment 2: Data Distribution

- Two data settings:
  - **IID**: each client has a random subset of digits
  - **Non-IID**: each client has only specific digits
- Compare:
  - Accuracy
  - Convergence behavior
- Key insight:
  - Why non-IID data is challenging for federated learning



# Optional Extensions (Choose One)

Depending on interest:

- Partial client participation per round
- Effect of local training epochs
- Communication-efficiency trade-offs
- High-level discussion of privacy benefits

These are **extensions**, not mandatory.

# What This Project Is NOT

- No new federated learning algorithms
- No heavy mathematical theory
- No cryptographic implementations
- Focus is on **understanding, experimentation, and explanation**

# Expected Thesis Structure

1. Introduction and motivation
2. Background on centralized and federated learning
3. System model and experimental setup
4. Experimental results
5. Discussion and insights
6. Conclusion

# Next Steps

Before the next meeting:

- Skim the provided papers at a high level
- Familiarize yourselves with the federated learning workflow
- Think about which optional extension interests you most

# Discussion

- Questions?
- Preferred tools or frameworks?
- Interests or ideas you would like to explore?